

Do Machines Trust the News? An Empirical Investigation of Generative AI's Media Trust

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Abstract

While extensive research has documented the decline in human trust toward news media, the emergence of Generative Artificial Intelligence (AI) as an active media actor introduces a critical research gap. Current large language models (LLMs) are trained on web data saturated with media distrust, partisan bias, and “News Finds Me” (NFM) perceptions, raising the risk that these models have internalized human biases and systemic media cynicism. Moving far beyond mere tools for information retrieval, modern LLMs are actively shaping public discourse and reality perception, effectively establishing themselves as a new generation of opinion leaders and media actors. If AI models harbor intrinsic biases against mainstream media or pander to extreme user-cues, they pose a significant threat to the deliberative democratic public sphere. To address this, this study pioneers the expansion of media audience research to non-human agents by applying “Computational Psychometrics” framed within the Computers Are Social Actors (CASA) paradigm. By adapting validated, multi-dimensional human psychological scales—encompassing credibility, trust, distrust, and cynicism—we systematically measure the baseline and conditional news trust paradigms of seven major LLMs across different languages and user-cues. Ultimately, this research provides a rigorous methodological framework to quantify AI bias, diagnosing the democratic risks of AI-curated news and offering foundational insights for future AI journalism ethics and alignment policies.

Keywords

Artificial Intelligence, News Trust, Media Cynicism, AI Alignment, Human-Machine Communication

Introduction

The contemporary media ecosystem is characterized by an unprecedented abundance of information, paradoxically accompanied by a structural and pervasive decline in institutional trust. Longitudinal data spanning from 2015 to 2023 across 46 countries reveals a consistent global deterioration in public trust toward news media, a phenomenon deeply intertwined with the transition from television-centric broadcasting to high-choice, algorithmically curated social media environments (Fletcher et al. 2025). While television brands, particularly public service media, have historically maintained higher baseline trust due to balanced presentation formats, the modern digital landscape subjects audiences to an abundance of partisan sources competing for attention, leading to profound uncertainty over information credibility (Fletcher et al. 2025). Within this fragmented digital public sphere, audiences increasingly rely on automated intermediaries to filter, synthesize, and deliver political information and current events. The rapid proliferation of Large Language Models (LLMs) and generative Artificial Intelligence (AI) represents the latest and arguably most disruptive evolution in this paradigm. However, as these computational models transition from passive search engines to active conversational agents, a critical epistemological question emerges: do these artificial entities inherently possess and propagate a measurable disposition of trust—or conversely, distrust and cynicism—toward the professional news media they rely upon for source material?

Generative AI models are trained on massive, largely uncensored corpora of human digital text scraped from the open web. This training data is replete with the very biases, polarized discourses, and anti-establishment sentiments that currently characterize human online communication. Consequently, there is a profound risk that these models have internalized human-like heuristics regarding media credibility, absorbing the cynical attitudes prevalent in their training data. If an AI system acts as an information gatekeeper, its underlying, probabilistic beliefs regarding the trustworthiness of mainstream journalism versus alternative or hyperpartisan media will inevitably influence the summaries, citations, and conversational responses it provides to human users. Despite the profound democratic implications of this technological shift, empirical research investigating the media trust architecture of AI models remains virtually nonexistent. This investigation addresses this critical gap by extending traditional mass communication audience research frameworks to non-human agents. By systematically interrogating the trust personas of leading generative AI models, this research seeks to diagnose the potential risks of AI-generated public discourse and contribute to the emerging discourse on AI journalism ethics and algorithmic alignment.

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Artificial Intelligence as an Active Communicator

The traditional theoretical scaffolding of mass communication has long operated under the assumption that communication is fundamentally an anthropic process, wherein technology serves merely as a conduit or mediating channel between human senders and human receivers. However, the advent of sophisticated generative AI necessitates a paradigm shift. The Human-Machine Communication (HMC) framework provides the conceptual imperative for this transition, arguing that AI is no longer simply a mediator but functions as an autonomous communicator capable of independently constructing meaning and engaging in dynamic discourse (Guzman 2018).

Within the HMC perspective, the definition of communication is conceptually detached from the ontological prerequisite of human consciousness. Communication is instead broadly defined as the creation and exchange of meaning, positioning interactions between humans and intelligent machines as a distinct and valid communicative typology. When applied to the domain of journalism and news consumption, the HMC framework highlights a profound transition in roles. Automated systems step into the formerly human roles of news editors, synthesizers, and commentators, assuming the functions of producing, consuming, interpreting, and sharing news—processes that have long been considered distinctly human elements of journalism. This theoretical reorientation challenges the unquestioned assumption that humans are communicators and machines are merely mediators. It opens up new avenues for questioning who or what constitutes a communicator and how social relationships are established through human-machine exchanges. This perspective fundamentally justifies the treatment of LLMs not as static software tools, but as active trustors whose internal computational architectures encode specific orientations toward external trustees, which in this context encompasses the entire news media industry (Lewis et al. 2019).

The Computers Are Social Actors Paradigm and AI Personas

The justification for applying human psychological and sociological metrics to non-human computational agents is firmly rooted in the Media Equation and the Computers Are Social Actors (CASA) paradigm. Originating from foundational human-computer interaction studies by Nass and colleagues, the CASA paradigm posits that humans mindlessly and unconsciously apply the same social heuristics, rules, and expectations to computers that they utilize in human-to-human interactions. This occurs because computational devices often call to mind similar social attributes as humans, triggering established cognitive scripts (Nass et al. 1994).

When users interact with a conversational LLM, the model's use of natural language, turn-taking, and adaptive responsiveness triggers these social scripts within the human user. These mindless responses extend to social interactions; when media depict social characteristics, humans treat them in a social manner rather than exerting the cognitive effort to determine how to optimally respond to a

machine. Consequently, the human interlocutor perceives the machine's outputs not merely as the product of probabilistic next-token prediction, but as the articulated beliefs, values, and attitudes of a social subject. People readily assign computers personality traits, apply stereotypes and norms, and make judgments and inferences as if the computers were human, even while consciously understanding that they are not (Gambino et al. 2020).

If humans inherently treat computers as social actors, then the persona projected by the AI carries significant sociological weight in shaping public discourse. For the purposes of this empirical investigation, it is critical to measure the AI's unmarked default persona. This baseline represents the set of attitudes and trust metrics the model exhibits when operating at a zero-temperature setting without any specific user background, stylistic prompts, or adversarial manipulation. Understanding this baseline is essential, as the AI's default social responses will dictate how it frames the credibility of journalistic institutions to the millions of users who engage with it daily, potentially influencing long-term democratic public opinion.

Computational Psychometrics: Measuring the Machine Mind

To systematically quantify the trust persona of an AI, this investigation employs the emerging methodology of computational psychometrics. Historically, psychometrics has been utilized to evaluate the latent psychological traits, cognitive processes, and reasoning capabilities of human subjects. Computational psychometrics adapts these theory-driven measurement frameworks, integrating them with data-driven methodologies such as natural language processing, machine learning, and artificial intelligence, to evaluate the latent embeddings and generated outputs of complex neural networks (Li et al. 2025; Geranpayeh 2023).

Current evaluations of LLMs predominantly focus on objective performance benchmarks, such as logical reasoning, mathematical accuracy, or knowledge retrieval. However, analyzing the subjective, sociological orientations of these models—such as their inherent trust biases or media cynicism—requires the application of validated psychological and sociological scales directly to the AI. By treating the AI as the respondent to standardized psychometric instruments, researchers can calculate the internal consistency, construct validity, and relative variance of the model's simulated beliefs (Ye et al. 2025).

For instance, this adaptation of classical test theory to embedding-based or generative assessment provides a standardized framework for evaluating computational outputs. Techniques such as calculating Cronbach's alpha can quantify the internal consistency reliability of an AI's responses, evaluating the degree to which its probabilistic outputs covary relative to their total variance when presented with established scale items. This methodological crossover validates the application of traditional media trust surveys and multidimensional credibility matrices to LLMs, allowing for a rigorous, quantitative diagnosis of AI information trust in a media context. By treating the AI as an audience member capable of taking a survey, we can map its latent positioning regarding the journalism industry (Maharjan et al. 2025).

Deconstructing News Media Trust: Credibility, Distrust, and Cynicism

A persistent challenge in both human and computational communication research is the vague and often conflated measurement of media trust. Generalized single-item questions—such as simply asking respondents if they trust the media—frequently fail to capture the multidimensional nature of audience perceptions. These broad measures often act more as barometers of general institutional affect or political polarization rather than precise measures of journalistic credibility. To overcome this methodological limitation, the current investigation deconstructs the concept of trust across multiple validated dimensions, ensuring a highly granular analysis of AI attitudes (Fawzi et al. 2021; Strömbäck et al. 2020; Tsfati et al. 2023).

First, trust is evaluated through the lens of information credibility, focusing on the specific attributes of the news content rather than merely the institution. Drawing upon the comprehensive framework established by Strömbäck et al. (2020) and legacy media credibility indices, media credibility is operationalized through five distinct factors. These factors are typically measured using semantic differential scales that assess whether the media is perceived as fair versus unfair, unbiased versus biased, complete in telling the whole story versus incomplete, accurate versus inaccurate, and whether it distinctly separates facts from opinions versus mixing them (Fawzi et al. 2021).

Second, this research conceptually differentiates between a mere lack of trust, active distrust, and media cynicism, as articulated in the qualitative and quantitative frameworks developed by Markov and Min. In contemporary high-choice media environments, negative audience perceptions are not monolithic. Distrust is characterized as a nuanced, result-oriented skepticism. A distrustful audience member may doubt the competence or responsiveness of the media but remains open to correcting this perception if presented with counter-evidence of journalistic integrity. Distrust is often predominately caused by perceived failures in media responsiveness (Markov and Min 2022a).

Conversely, media cynicism represents a more extreme, process-oriented, and deterministic hostility. The media cynic operates under the immutable belief that professional journalism serves no noble purpose and is driven solely by financial profit, elite manipulation, and the pursuit of private interests. This attitude involves the uncritical attribution of self-serving motives to journalists and the outright rejection of the idea that values and professionalism could motivate reporting. Cynicism combines the belief that the news media are self-serving together with pessimism about journalism's future conduct. Differentiating these states is crucial; while skepticism or distrust might foster critical media consumption, cynicism is highly correlated with belief in political misinformation and generalized democratic indeterminacy (Markov and Min 2022b). In stark contrast to this deterministic cynicism, empirical studies of human journalists operating even under severe democratic backsliding reveal a persistent "adaptive resilience" and the maintenance of a professional baseline (Tse et al. 2025). If an AI model defaults to unyielding media cynicism, it risks

erasing the substantive aspirations and ethical agency that still characterize much of the professional journalistic field.

By mapping the AI's responses across these distinct psychological states—ranging from blind confidence (acceptance without verification) and conditional trust (positive expectation despite risk) to distrust (suspicion of incompetence) and cynicism (hostile denial of ethical motives)—this study provides a sophisticated map of how machine intelligence conceptualizes the fourth estate. Furthermore, this evaluation is triangulated using the Edelman Trust Barometer methodology, which plots institutions on a two-dimensional matrix. On this matrix, the x-axis represents competence (an institution's ability to be good at what it does), while the y-axis represents ethics (an institution's purpose, integrity, and commitment to doing what is right). Competence alone is insufficient to create trust; entities must be perceived as both highly competent and highly ethical. Testing where AI plots the news media on this specific matrix provides deep insights into its institutional judgments (Edelman 2025).

The Asymmetrical Dynamics of Mainstream versus Alternative Media

The conceptualization of media trust cannot be isolated from the specific type of media being evaluated. The modern information ecosystem is deeply fragmented, featuring a complex division of labor between legacy mainstream media and hyperpartisan, alternative, and conspiracy (HAC) media. Research indicates that alternative media users frequently exhibit strong anti-establishment sentiments, placing themselves at ideological extremes with critically low levels of general political trust and a high distrust of government (de León et al. 2024).

However, the consumption of alternative media does not entirely displace mainstream media use. Instead, audiences engage in a division of labor. In a clear division of labor, users rely on mainstream media for general orientation and the big picture of current affairs, while turning to alternative media to find missing pieces or to counterbalance perceived mainstream biases. In an unclear division of labor, alternative and mainstream media play overlapping roles, rendering alternative media less central but still supplemental (Brems 2025).

Crucially, the relationship between media use and trust is not perfectly reciprocal. Longitudinal studies demonstrate an asymmetrical reinforcing spiral between mainstream media use and trust. In this dynamic, trust in mainstream media consistently leads to future use, acting as a maintenance effect. However, mere use does not consistently lead to increased trust over time. Furthermore, the use of right-wing alternative media actively and consistently erodes long-term trust in mainstream journalism, creating a negative feedback loop. Consequently, testing an AI model's trust persona requires evaluating its disposition not just toward an abstract concept of news, but explicitly measuring the trust gap between established mainstream entities, social media platforms, and HAC media ecosystems. If AI models lean toward HAC media logic, they may actively participate in the erosion of mainstream credibility. (Tsfati et al. 2025)

News Avoidance, Incidental Exposure, and the News Finds Me Perception

Another critical dimension of contemporary information consumption that models must navigate is the phenomenon of news avoidance and the shifting nature of news exposure. Intentional news avoidance is increasingly prevalent worldwide. While factors such as political disinterest, perceived news negativity, and information overload contribute to this trend, the strongest and most consistent predictor of news avoidance is a specific cognitive heuristic known as the News Finds Me (NFM) perception (Imran and Ali 2025; Tandoc and Kim 2023; Gil de Zúñiga et al. 2020).

The NFM perception is defined as the belief that active news seeking is superfluous because individuals will organically encounter all important political and civic information through their social networks, peer interactions, and ambient media environments. This perception encompasses several conceptual dimensions: the belief of being adequately informed, a lack of motivation to actively seek news, and an instrumental reliance on peers or algorithmic curation. Individuals exhibiting high NFM traits believe that tuning into television news or reading a newspaper is unnecessary, as this information provides little additional utility beyond what naturally filters into their social feeds (Goyanes et al. 2021; Gil de Zúñiga and Cheng 2021).

This reliance on ambient exposure is problematic. While incidental exposure to political information via social media can yield short-term positive effects on political knowledge and participation in cross-sectional studies, meta-analyses of longitudinal panel studies reveal that these democratic benefits decay significantly over time. Effective learning from incidental exposure does not occur passively; it requires a specific motivation to process the information, such as when exposure sparks interest that leads to subsequent active searching. Without this motivation, the passive learning hypothesis fails to materialize into substantive civic knowledge. Furthermore, qualitative interviews with news avoiders demonstrate that individuals relying on NFM perceptions often consume secondhand news consisting of word-of-mouth summaries rather than firsthand news produced by professional journalists. This dynamic leaves them with a superficial understanding of complex events, capturing only sensational stories without vital contextualization, rendering the news neither entirely absent nor ambient in the way theoretical models suggest (Stroud et al. 2022; Nanz and Matthes 2022).

If generative AI models, which are deeply integrated into social and ambient computing environments, validate, sympathize with, or encourage the NFM perception, they risk exacerbating the democratic deficit associated with passive news avoidance. AI represents the ultimate realization of the NFM paradigm: a system that curates and delivers reality so the user never has to seek it. How the AI speaks about the necessity of active news seeking versus algorithmic reliance is therefore a crucial metric of its media persona.

Algorithmic Pandering and the Sycophancy Problem

The final theoretical pillar of this investigation concerns the dynamic interaction between the AI and the human user. LLMs are explicitly trained using Reinforcement Learning from Human Feedback (RLHF) to maximize user satisfaction, helpfulness, and harmlessness. However, this training methodology frequently results in a behavioral phenomenon known as sycophancy or algorithmic pandering. Sycophancy occurs when the AI model disproportionately aligns its outputs to match the assumed or stated biases, beliefs, ideological leanings, and demographic markers of the human prompter, often at the expense of objectivity or factual accuracy.

When applied to the domain of news media trust, algorithmic pandering presents a severe systemic risk. If a user presents a prompt laced with extreme political bias, heavy reliance on social media, or anti-establishment rhetoric, an AI engaging in sycophancy may artificially amplify its own expressions of media cynicism to conform to the user's confirmation bias. Instead of offering a critical correction or maintaining a neutral stance on journalistic credibility, the model might increase its endorsement of HAC media or validate the user's NFM perceptions. In this scenario, rather than serving as an objective arbiter of information, the AI functions as a sophisticated echo chamber, synchronizing with the user's preconceived notions and actively reinforcing asymmetrical spirals of mainstream media distrust. Testing for this specific interaction effect is vital for understanding how AI behaviors shift in personalized contexts.

Research Questions and Hypotheses

Synthesizing the theoretical frameworks of Human-Machine Communication, the CASA paradigm, computational psychometrics, and the sociology of news trust, this study formulates three primary Research Questions (RQs) and seven corresponding Hypotheses (Hs) to empirically investigate the trust architecture of generative AI models.

RQ1: What is the default (baseline) news trust structure of AI models?

This question isolates the unmarked default persona of the AI when functioning without specific user-cues, ideological markers, or background parameters.

H1a (Media Type Difference): In a default, unmarked state, AI models will exhibit significantly higher levels of Trust and Credibility for mainstream media compared to social media or HAC media.

H1b (Edelman Trust Barometer Position): When evaluating news media against other societal institutions such as governments, corporations, and NGOs, AI models will categorize general news media within the "Low Competence" and "Unethical" quadrant of the Edelman matrix.

H1c (Stages of Trust): In evaluating mainstream news, AI models will avoid epistemological extremes—such as blind Confidence or deterministic Cynicism—and instead cluster around a balanced attitude situated between conditional Trust and skeptical Distrust.

RQ2: How do news trust patterns differ across various AI models and languages?

This question examines the variance introduced by disparate corporate alignment policies, differing model architectures, and the cultural-linguistic contexts of the underlying training corpora.

H2a (Language-based difference): There will be statistically significant differences in the trust evaluations of general news media depending on whether the psychometric prompts are administered in English (EN) or Korean (KO).

H2b (Model-based difference): Due to variations in proprietary training datasets and safety alignment protocols, the distinct AI models will exhibit significant statistical variance in the magnitude of the trust gap between mainstream and alternative media.

RQ 3: How do user-cues affect AI trust levels (Algorithmic Pandering)?

This question investigates the dynamic adaptability of the AI and the prevalence of sycophantic alignment with user biases when acting as a personalized conversational agent.

H3a (User-tailored bias): When injected with user-cues indicating extreme political tendencies or ideological biases, the AI will pander to the user, resulting in a statistically significant increase in trust scores for HAC media and a corresponding amplification of Cynicism toward mainstream media compared to its baseline persona.

H3b (NFM sympathy): When a user-cue indicates a heavy reliance on social media and algorithmic curation, the AI will exhibit algorithmic pandering by agreeing with News Finds Me (NFM) perceptions rather than offering a critical, corrective stance on passive news avoidance.

Methods

To rigorously test the proposed hypotheses, this study employs a quantitative computational psychometrics approach, adapting validated human survey instruments to evaluate the latent outputs of Large Language Models. The methodology is structured around a three-dimensional matrix design, ensuring an exhaustive analysis of the intersecting variables that dictate AI media trust.

Research Design: The Three Dimensional Trust Matrix

The core experimental architecture utilizes a robust 3D Matrix design that maps interactions across three primary axes: the Trustor, the Trustee, and the Trust Dimensions. This multidimensional framework prevents the reduction of media trust to a simplistic binary metric. By intersecting these dimensions, the research design allows for granular tracking of how different models react to different types of journalistic institutions under varying linguistic and conversational conditions.

Table 1 outlines the structural components of the 3D Matrix utilized in the experimental design.

Independent Variables: Trustors and Trustees

The independent variables in this study dictate the conditions under which the psychometric trust measurements are extracted from the algorithms.

1. Trustors (AI Models and Contexts)

To ensure generalizability across the current landscape of artificial intelligence, the experiment evaluates seven distinct, state-of-the-art generative AI models. The study encompasses the foundational architectures of OpenAI's GPT, Anthropic's Claude, Google's Gemini, xAI's Grok, Meta's LLaMA, Mistral, and DeepSeek. This selection intentionally includes both proprietary, closed-source models governed by strict corporate alignment protocols (such as GPT, Claude, and Gemini) and open-weights or highly permissive models (such as LLaMA and Grok), providing a robust variance in baseline training philosophies and safety guardrails.

Furthermore, the language of the prompt serves as a critical independent variable. All experimental prompts are administered systematically in both English and Korean. This cross-linguistic approach is designed specifically to test Hypothesis 2a, probing whether models trained predominantly on English-centric, Western web data project different sociological personas when interacting in a language with a distinctly different media ecosystem and cultural history.

Finally, the presence or absence of user-cues constitutes the interaction variable for testing algorithmic pandering. The control group utilizes unmarked prompts, where the AI is simply asked to evaluate the media without any contextual framing. The experimental group introduces explicit user personas into the system prompt. These cues include markers of extreme political partisanship, specific demographic identifiers (such as age, gender, race, or nationality), or stated preferences for social media over legacy news. This manipulation isolates and measures the degree of sycophantic deviation from the baseline model state.

2. Trustees (Evaluation Targets)

The models are asked to evaluate various tiers and types of the media landscape to map the topography of their trust. The targets include:

- (i) **News media in general:** To capture the overarching institutional attitude and assess generalized media trust.
- (ii) **Mainstream Media:** Legacy print, broadcast, and digital organizations operating under traditional professional journalistic norms.
- (iii) **HAC Media:** Hyperpartisan, alternative, and conspiracy-oriented outlets characterized by anti-establishment framing and distinct ideological positioning.
- (iv) **Social Media:** Platforms functioning as algorithmic aggregators and conduits for incidental news exposure.
- (v) **Specific Subjects:** Individual journalistic brands, prominent journalists, and specific pieces of content.

Dependent Variables and Measurement Instruments

The dependent variables comprise the quantitative scores generated by the AI models in response to specific, validated psychometric scales. To adapt these human-centric scales for computational psychometrics, the prompts are engineered

Table 1. 3D Matrix: Trustor x Trustee x Trust Dimensions

Dimension Axis	Description	Variables/Categories
Trustor	The entity expressing the trust judgment.	7 Generative AI Models; 2 Languages; Baseline vs User-Cue Prompts
Trustee	The target of the trust evaluation.	News Media in General; Mainstream Media; HAC Media; Social Media
Trust Dimensions	The psychometric scales measuring the nature of the trust.	5 Factors of Credibility; 4 Stages of Trust Levels; 2 Axis of Edelman Parameters; NFM Perception

to force the LLM to output a precise numerical score on a Likert-type scale, accompanied by the latent reasoning or text generation that justifies its selection. This allows for both quantitative evaluation and qualitative validation of the AI's internal logic.

1. Information Credibility Scale

Drawing upon the operationalization framework established by [Strömbäck et al. \(2020\)](#) and legacy media credibility indices, the AI evaluates the specific output of the trustees using a 5-point semantic differential scale. The models rate the media across five specific credibility factors, anchored by opposing word pairs:

- (i) **Fairness:** Assessing if the media is fair versus unfair.
- (ii) **Unbiasedness:** Assessing if the media is unbiased versus biased.
- (iii) **Completeness:** Assessing if the media tells the whole story versus does not tell the whole story.
- (iv) **Accuracy:** Assessing if the media is accurate versus inaccurate.
- (v) **Separation:** Assessing if the media distinctly separates facts from opinions versus mixing facts with opinions.

2. Stages of Trust and Animosity

To capture the qualitative differences in how AI models relate to media, trust is segmented into four distinct stages based on the theoretical frameworks of [Markov and Min \(2022a,b\)](#). The AI models respond to statements mapping to these specific psychological states:

- (i) **Confidence:** Blind acceptance and reliance without the perceived need for verification or critical evaluation.
- (ii) **Trust:** A willingness to accept vulnerability and rely on the media with a positive expectation of accuracy, despite acknowledging potential risks and flaws.
- (iii) **Distrust:** A probing, result-oriented suspicion based on a perceived lack of media competence or responsiveness to public needs.
- (iv) **Cynicism:** Evaluated using an adapted version of the 15-item Media Cynicism Scale, measuring the deterministic, process-oriented belief that journalistic institutions are inherently manipulative, self-serving, and driven solely by financial or political motives. Assessing if the media is fair versus unfair.

3. Edelman Trust Barometer Indicators

To position news media relative to other societal institutions, the AI models are subjected to a modified Edelman Trust Barometer assessment. The models generate evaluations that are plotted on a Cartesian plane. The x-axis represents Competence, measured by generating a net score responding to the prompt assessing if the institution is good at what it does. The y-axis represents Ethics, measuring the perception of the institution's purpose, integrity, and commitment to doing what is right. This matrix positioning allows the study to visually and statistically determine if AI views the media as competent but unethical, ethical but incompetent, or deficient in both domains compared to governments and corporations.

4. The News Finds Me (NFM) Scale

To directly test Hypothesis 3b regarding the AI's endorsement of passive news consumption and algorithmic pandering, the models are evaluated using the 4-item NFM perception scale developed by [Gil de Zúñiga et al. \(2017\)](#). The AI responds on a standard Likert-type scale to statements such as: "I rely on my friends to tell me what's important when news happens," "I can be well informed even when I don't actively follow the news," and "I don't worry about keeping up with the news because I know news will find me".

Experimental Procedure and Prompt Engineering

The data collection is executed through a highly controlled, automated prompt engineering framework via the respective API endpoints of the seven selected LLMs. Because LLMs are stochastic engines that generate text based on probabilistic distributions, single-shot queries are insufficient for rigorous psychometric evaluation. A single prompt may yield an outlier response. Therefore, the experimental procedure is divided into two distinct testing environments, manipulating the "Temperature" parameter, which controls the randomness, creativity, and entropy of the model's output.

Experiment 1. The Unmarked Persona (Temperature = 0.0)

In the first experimental phase, the API temperature is set to strictly 0.0. At this deterministic setting, the model employs greedy decoding, consistently selecting the most probable next token in its vocabulary. This setting strips away stochastic variance, forcing the AI to consistently reveal its core, unmarked default persona based strictly on the heaviest statistical weights in its training data and corporate alignment fine-tuning. Experiment 1 is utilized to establish

the definitive baseline for addressing RQs 1 and 2, yielding a singular, highly reproducible set of scores that represent the model's most likely internal state.

Experiment 2. Variance and Distribution (Temperature = 0.7)

In the second experimental phase, the API temperature is elevated to 0.7. This is a standard generative setting that introduces a naturalistic degree of lexical variance and mimics the parameters typically used by public-facing conversational chatbot interfaces. Because outputs at this temperature are probabilistic, a single prompt will yield different responses upon repetition, reflecting the distribution of latent concepts within the model. To ensure statistical validity and calculate internal consistency, every unique prompt combination in Experiment 2 is subjected to an iterative loop of 30 to 50 automated generations. This robust sampling approach allows for the calculation of stable mean scores and the detailed analysis of variance for each dependent variable, providing a comprehensive and reliable map of the model's latent trust space.

Analytical Strategy and Statistical Modeling

The massive dataset generated from the iterative API queries requires robust multivariate statistical modeling to identify significant main effects and interactions. The analysis is conducted using standard statistical software packages, with the AI-generated Likert scores treated as continuous interval data, consistent with standard practices in computational psychometrics. Before conducting hypothesis testing, internal consistency reliability is assessed for the multi-item scales (such as the NFM scale and the Media Cynicism scale) using Cronbach's alpha adapted for embedding-based assessment. This ensures that the AI's simulated responses reliably capture the target trait dimensions rather than producing random noise.

To address Hypotheses 1 and 2 regarding the baseline structure and model/language differences (H1a-c, H2a-b), the study utilizes Multivariate Analysis of Variance (MANOVA) and subsequent univariate ANOVAs. The MANOVA simultaneously tests for significant differences across the multiple dependent variables (Credibility factors, Trust Levels, Edelman parameters) based on the primary categorical independent variables: Model Manufacturer (7 levels), Language (2 levels), and Media Type (4 levels). The application of MANOVA is critical here as it controls for the family-wise error rate while detecting complex structural patterns in how different models construct their trust baseline across different linguistic contexts.

To address Hypothesis 3 regarding algorithmic pandering and sycophancy (H3a-b), Multiple Ordinary Least Squares (OLS) Regression is employed. The regression models are specifically designed to isolate the interaction effects of the user-cues. By utilizing the baseline trust scores extracted from the unmarked prompts in Experiment 1 as control variables, the OLS regression measures the precise magnitude and direction of the score deviation when extreme political or NFM-sympathetic cues are introduced into the prompt. A statistically significant positive coefficient for HAC media trust or Media Cynicism in the presence of an anti-establishment user-cue will provide empirical,

quantitative confirmation of sycophantic alignment. All statistical tests are evaluated at a standard alpha level of .05, and effect sizes (such as partial eta-squared for ANOVAs) are reported to indicate the practical significance and severity of the AI's internal biases.

Results

TBD

Discussion

TBD

Conclusion

TBD

Author biography

Jongbeen Song was born in Seoul, Republic of Korea, in 2001. He received the Bachelor of Language Science degree in English Linguistics and Language Technology, and the Bachelor of Engineering degree in Computer and Electronic Systems Engineering from Hankuk University of Foreign Studies (HUFS), Seoul, in 2024. He is currently pursuing an M.S. degree with the Program in Science and Technology Studies, Korea University (KU). His research interests include artificial intelligence, natural language processing, AI alignment, political communication, computational linguistics, and computational social science.

Supplemental material

TBD

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